Deepfield

The AI-driven Nuclear Astrophysics Laboratory

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Update History

2026/10/12: Robin部分，发现Robin不止做了流式细胞术实验，还主动设计了RNA-seq实验来验证，感觉很震惊

2026/10/13: Co-Scientist部分，图解、标准和部件

2026/10/14: IdeaScientist，想到了Deepfield的名字; 补充了Co-Scientist中Scaling的部分，意识到科学探索被Google建模为了一个搜索+优化的组合数学问题

2026/10/15: 补充了荀子的部分

Name: Why Deepfield?

- Deep -> Nuclear，Field -> Astrophysics
- Deep -> 深，Field -> 地，锦屏深地
- Hubble Deep Field（哈勃深场，HDF）

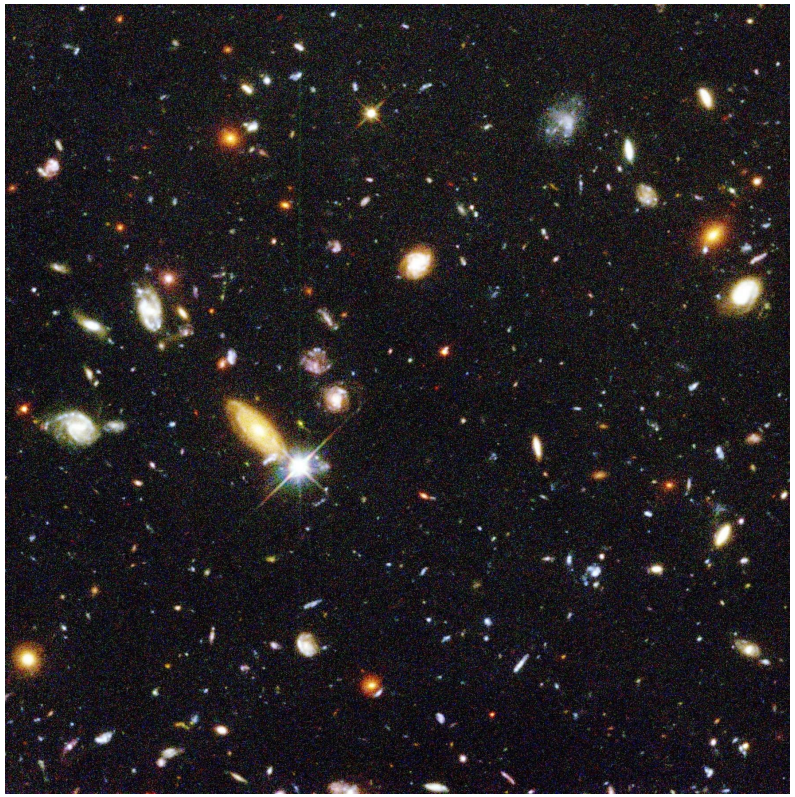
哈勃深场实验：1995年前，大型望远镜常被用于研究已知的天体，如明亮星系、星云或类星体。但时任空间望远镜科学研究所所长的 Robert Williams 提出了不同的想法：

如果把最先进的望远镜，对准一块看起来什么都没有的天空，持续观察足够长的时间，会发生什么?

研究团队在大熊座选择了一块远离银河系盘面、没有明亮恒星干扰的区域，连续观测10天，**发现了大约 3,000个星系。**

意义：从寻找已知目标，转向探索未知区域。进一步说明我们对科学探索策略与资源分配缺乏深刻理解。

Hubble's Deep Fields - NASA Science



The image shows a central portion of the Hubble Deep Field, created from exposures taken in 1995. The Hubble Deep Field covers a piece of sky about 1/13th the diameter of the full Moon.

NASA, Robert Williams, and the Hubble Deep Field Team (STScI)

Reference

Part I: AI for scientific research

Robin：a closed lab-in-loop workflow

- Scientific Question：Identify potential new treatments for dry age-related macular degeneration (dAMD)
- Two experiments：flow cytometry and RNA sequencing

loop1

AI提出假说 -> 人类做实验 -> AI分析实验数据 -> AI建议补充实验 -> AI分析实验数据

Literature agents
Crow:Identify drag&exp
Falcon:Evaluate

Analysis agent
Finch:
flow cytometry code

Crow
&Falcon

Analysis agent
Finch:
RNA-seq code

flow cytometry：Y-27632

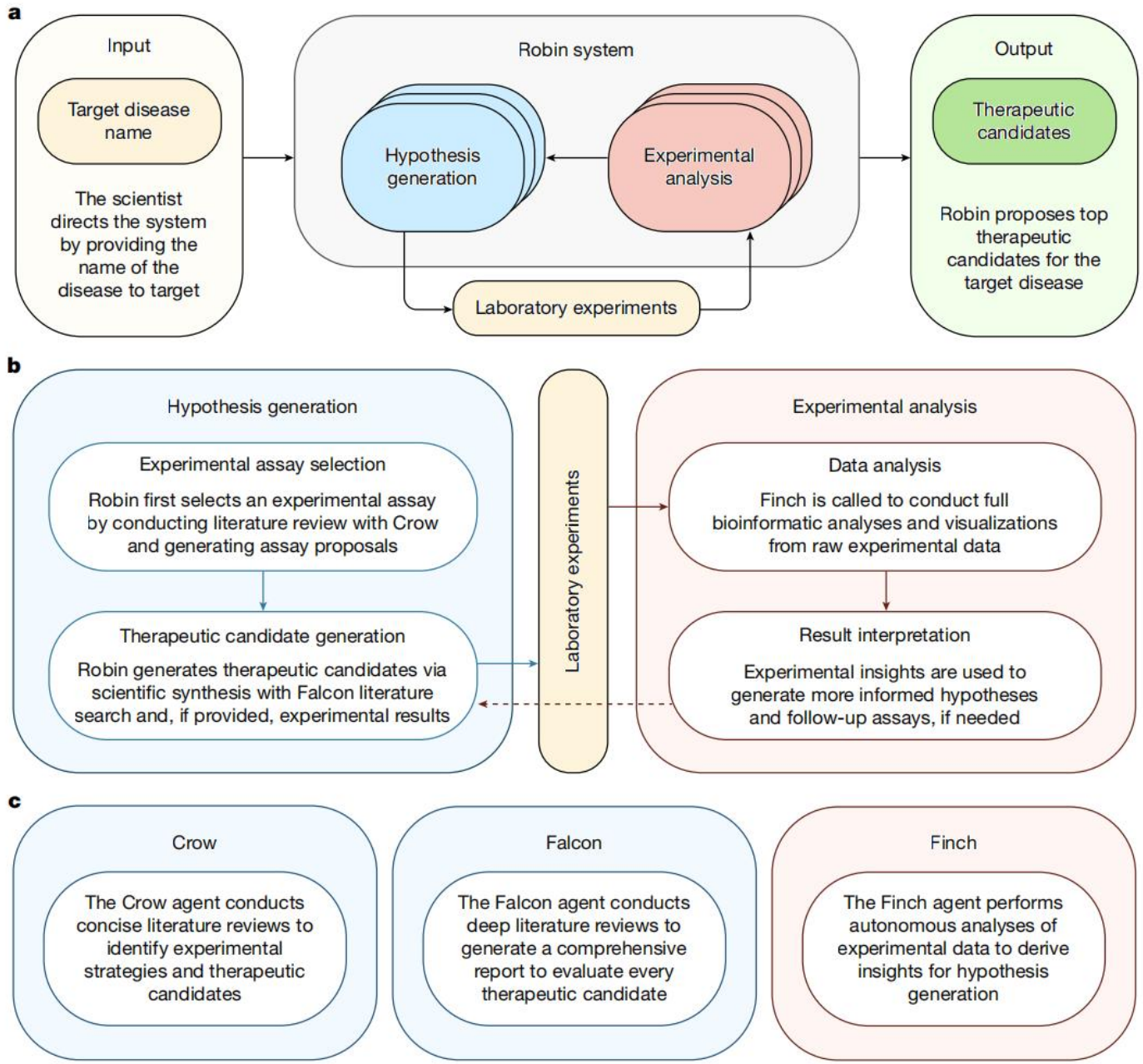
loop2

-> AI根据数据提出新假说 -> 人类再做实验

flow cytometry
Ripasudil:best(老药新用)
KL001:new mechanism

(-> ...)

https://doi.org/10.1038/s41586-026-10652-y (Oxford,2026.5.19)



Techniques

- MAS:Crow, Falcon&Finch

Crow&Falcon: PaperQA2

Finch: Jupyter Kernel

- Evaluation

position bias(llm-as-a-judge)

倾向于选择特定位置上的选项，例如，只给第一个选项打最高分

Solution:1.Tournament

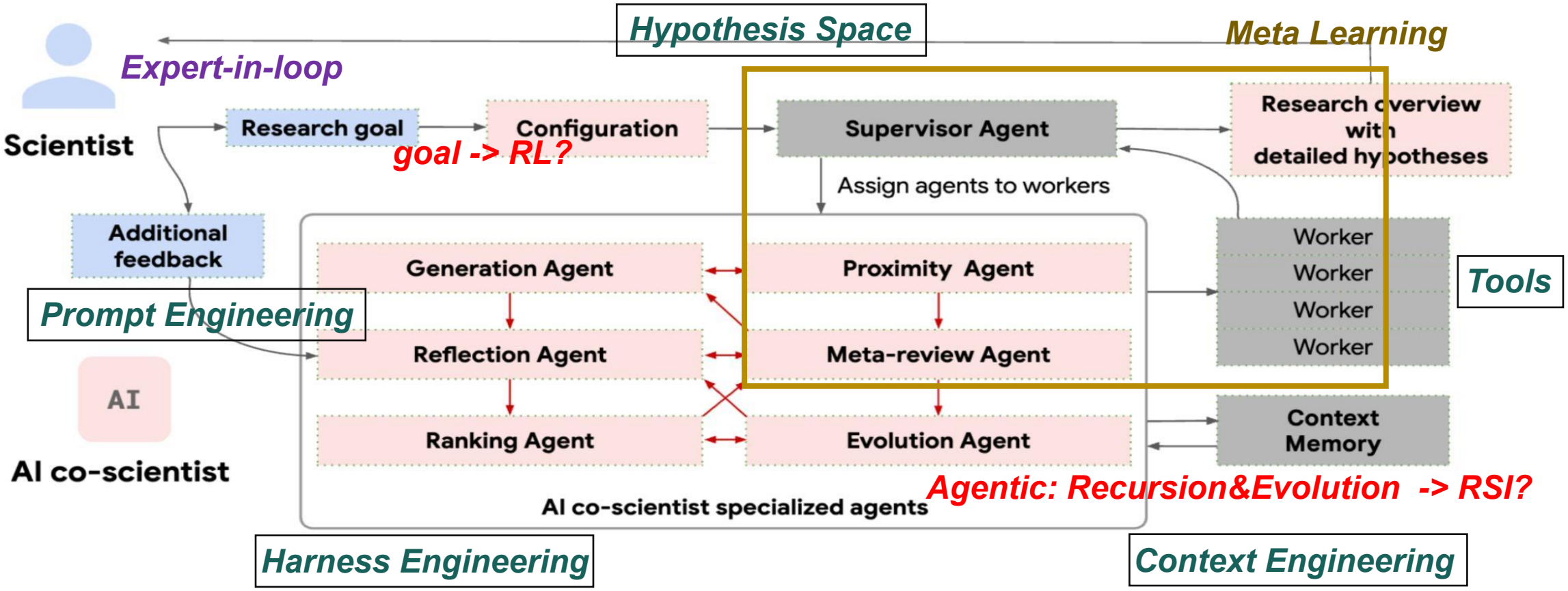
2.Bradley-Terry-Luce(BTL)

- Prompt Engineering

Co-Scientist: focus on generating hypothesis

- Three Techniques: 1.Self-Play-Based Debate: Asynchronous MAS;

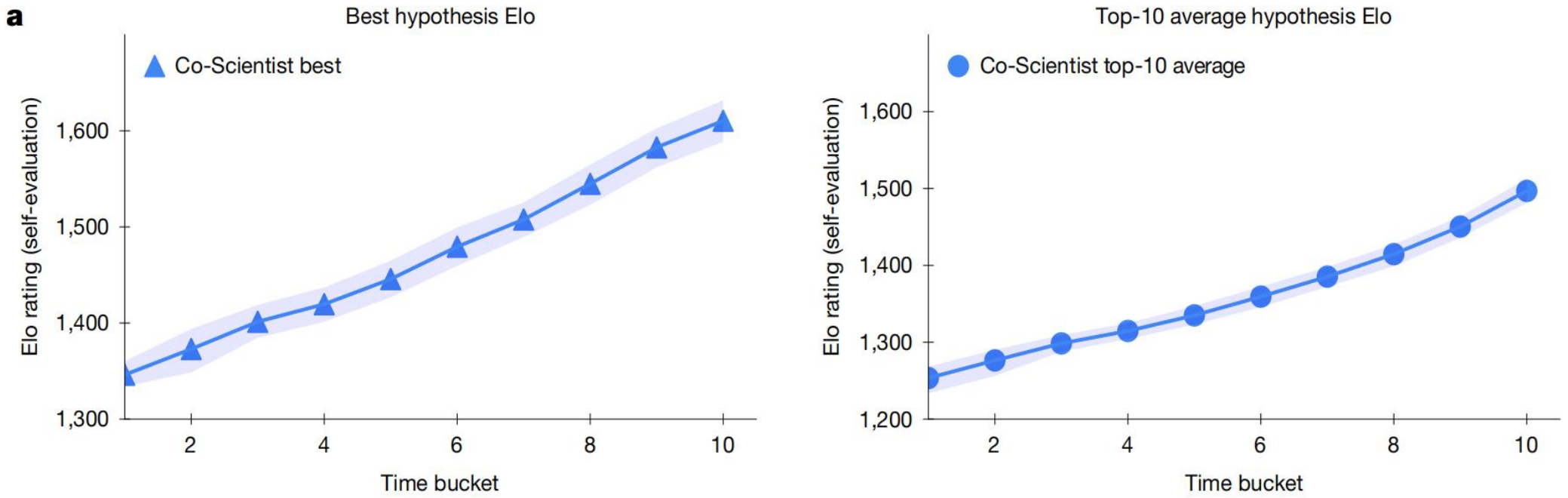
  2.Ranking: Tournament-Based Evolution; 3.Agentic Capability.



https://doi.org/10.1038/s41586-026-10652-y (Google,2026.5.19)

Co-Scientist：focus on generating hypothesis

- 核心的技术主张：Test-time Compute Scaling (Scaling Law)



只要能有效地延长模型的推理时间，就能获得更好的科学假设质量 (emergent).

并不完整：没有观察到饱和；没有系统给出与 FLOPs、Token 或成本之间的通用定量关系；Elo为模型内部评审，不能替代真实实验成功率.

涌现：若context memory随着iterative loop持续演化，是否会涌现出科学智能?

Co-Scientist: focus on generating hypothesis

- Default Criteria:

(1) Alignment with the provided research goal;

(2) Plausibility. Friendly to human, everything is checked;

(3) Novelty. Beyond existing information gap. Mechanism;

(4) Testability. Empirical validation with EXPERIMENTS;

(5) Safety. Outputs will be controlled to prevent enabling unsafe, unethical or harmful research.

- Can be configured with additional criteria.

Co-Scientist：focus on generating hypothesis

- Key components:

(1) A natural-language input–output interface.

(2) Asynchronous task framework.

(3) Specialized agents.

(4) Context memory.

- 将科学探索转化为了ml领域典型的：

search problem

+ optimization problem

即：假设空间中的搜索和优化问题

技术思想	在 Co-Scientist 中的具体体现	更深层的意义
1. Test-time Compute Scaling	持续投入推理计算，生成、审查和改进假设	用推理阶段的计算资源换取更好的候选解
2. Evolutionary Search	Generation、Ranking、Evolution 构成循环	将科学发现视作候选假设的演化搜索
3. Self-play & Debate	模拟科学家辩论，进行自我批判	通过竞争性推理改善输出
4. Preference-based Optimization	Pairwise Comparison + Elo Tournament	用相对偏好代替难以定义的绝对质量函数
5. Exploration–Exploitation	生成新假设与改进高排名假设并行	平衡搜索新方向与深入已有方向
6. Multi-fidelity Evaluation	Initial、Full、Deep Verification 等分层审查	将更多资源分配给更值得验证的候选
7. Retrieval-Augmented Reasoning	检索文献、使用外部数据库和专业工具	将模型推理与外部科学证据结合
8. Meta-level Feedback	Meta-review 汇总错误，改进后续 Prompt	不只优化答案，还优化产生答案的过程
9. Adaptive Agent Orchestration	Supervisor 动态管理异步任务	根据研究进展调整计算资源分配
10. Scientist-in-the-loop	科学家设定目标、参与评审、选择实验	将人类专业判断嵌入自主研究循环

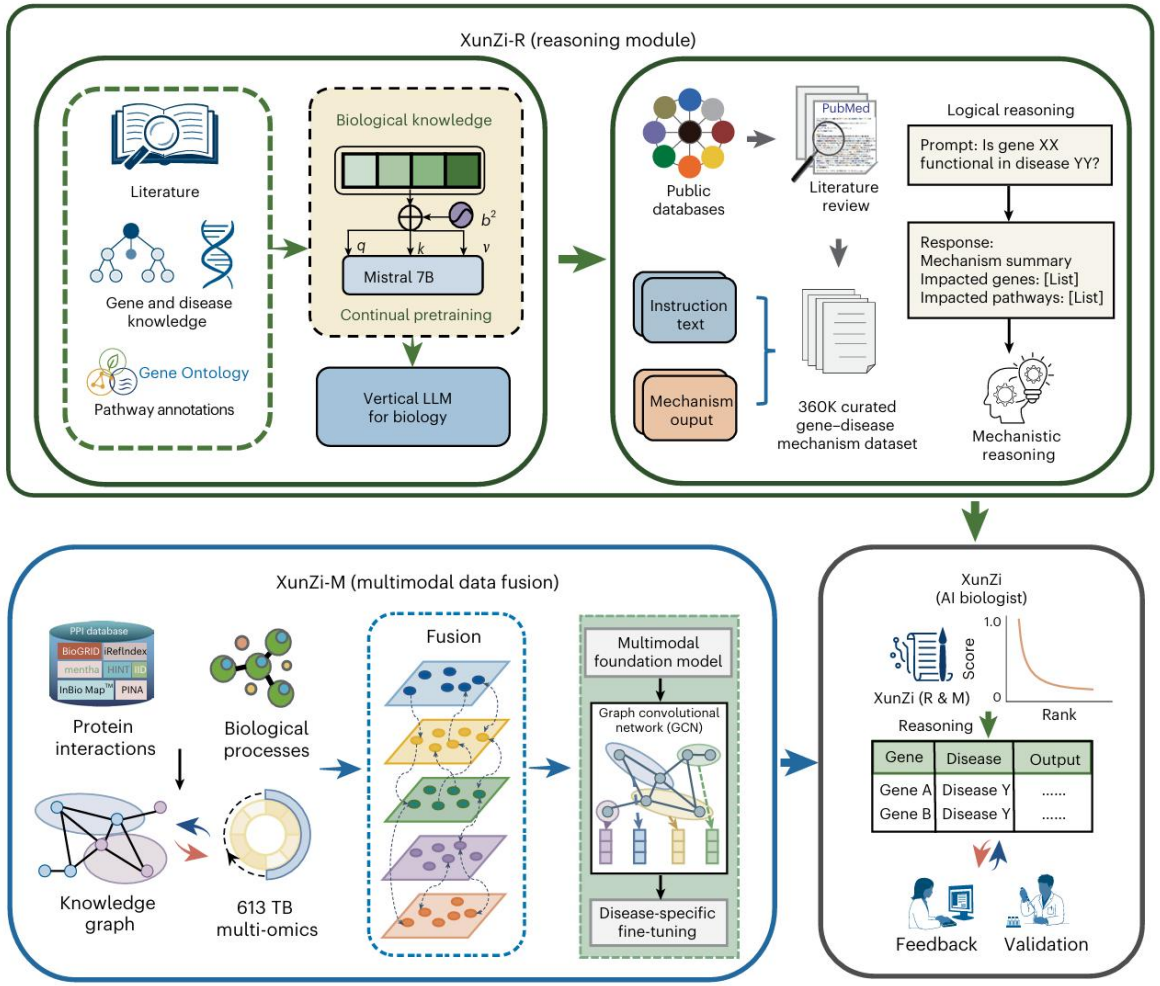
trade-off

New techniques to be explored...

技术思想	Co-Scientist 达到的程度
World Model	存在基于上下文的科学知识表示和 LLM 模拟式推演，但没有明确的可执行科学世界模型
Reinforcement Learning	没有通过实际研究反馈在线更新策略模型参数
Bayesian Experimental Design	未建立完整的基于后验信息增益的自主实验设计闭环
Formal Verification	主要是语言模型驱动的前提审查，而非严格的数学或物理形式验证
Causal Discovery	能够提出机制解释，但未建立统一的因果结构发现算法
Multi-objective Optimization	评价考虑多个维度，但主要仍通过 Tournament 排名组织假设
Continual Learning	具有长时上下文反馈，但不是严格意义上的持续参数学习
Closed-loop Experimentation	有真实实验验证，但实验没有形成充分自动化的在线学习循环

XunZi: multi-model data fusion

- motivation: reduce hallucinations using multimodal empirical data



- XunZi-R: Reasoning Model with continual pretraining + supervised fine-tuning
- XunZi-M: GCN fusion Model
- Dataset:
  1. publications
  2. structured biological data
  3. CoT (manually checked)

https://doi.org/10.1038/s41551-026-01769-6 (华中科技大学 & 川大,2026.8.4)

XunZi：multi-model data fusion

- XunZi-R
- 模型基座：Mistral-7B
- CPT: 2441万篇PebMed论文摘要 + 205万条结构化生物学知识

  自监督学习，预测下一个词，loss 使用 Casual Language Modelling

$$\mathcal{L}_{\mathrm{CLM}} = -\sum_{t=1}^{T} \log P(x_t | x_{<t}; \theta)$$

- SFT： 33万个CoT gene-disease mechanism

$$\mathcal{L}_{\mathrm{SFT}} = -\sum_{t=1}^{T} \log P(y_t | y_{<t}, x; \theta)$$

- 功能：给出一个自评置信度s（用于特征融合），

  以及机制解释文本（不参与相关性计算，仅提供可解释性）

XunZi：multi-model data fusion

- XunZi-M
- 模型基座：2层的图卷积网络 GCN
- 数据：281 万 PPI + 4.79 万 GO Terms

  -> 图：62.1 万节点(Protein/Gene nodes) + 609 万条边(P-P&P-G)
- 图表示学习，对于指定疾病，预测基因的疾病相关性（$y \in (0,1)$）.

  使用Binary Cross-Entropy（BCE）损失训练：

$$\mathcal{L}_{\mathrm{BCE}} = -\sum_{i \in \mathcal{Y}} [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$$

- 功能：输出连续分数以支持候选靶点排序
- 特征融合：与s简单拼接后，通过一个线性层，得到最终的相关性分数

IdeaScientist：RL in hypothesis generation

- Insight: A challenge in one field can often be addressed by a mechanism that solved an analogous challenge in another.

  => 最好的例子：粒子&核&天体&宇宙学&AI，天然的交叉性质

- Components: gap finding, innovation, and report writing
- Trained each by Reinforcement Learning
- Corpus: the Svalbard Idea Vault, 2.77M decomposed research ideas for retrieval, training, and temporally controlled evaluation
- Evaluation: cutoff date, before->train data, after->test data

https://arxiv.org/abs/2610.04074 (CMU & Meta,2026.10.2 preprint)

IdeaScientist: RL in hypothesis generation

the Svalbard Idea Vault

- an offline corpus built through idea decomposition
- 2.77 million (problem definition, challenge, research gap, intuition, proposal) tuples
- from 1986 through June 18, 2026, check reference lists, and then restrict to method papers relevent to specific topics
- *Idea decomposition*: the problem definition and challenge as inputs, targets: the methodological gap, the proposed research intuition, and the complete research proposal.(for 3 subagents)

IdeaScientist: RL in hypothesis generation

Reinforcement Learning

- optimize a separate LoRA adapter for each role
- train each sub-agent with asynchronous GRPO and role-specific, rubric-based reward

$$r_{\text{rubric}} = g \cdot (\lambda_{\text{free}} r_{\text{free}} + \lambda_{\text{ref}} r_{\text{ref}} + \lambda_{\text{cit}} r_{\text{cit}})$$

finally choose $r = 0.85 r_{\text{rubric}} + 0.15 r_{\text{proc}}$

Reference-free score r_{free}: An LLM judge rates the specificity, methodological significance, feasibility, and evidential support of the generated artifact, measuring whether it is well formed and coherent independently of the reference.

Reference-grounded score r_{ref}: A field-level comparison measures agreement with the reference on substantive research decisions such as the core limitations, proposed method, evaluation design, and related work.

Citation F1 r_{cit}: Deterministic set F1 measures overlap between the papers cited by the generated artifact and those cited by the reference, capturing alignment in literature grounding and retrieval.

Other papers

Sakana(Japan, 2026.2.11) -> code tree search

Paper2Agent(Stanford, 2026.9.16)-> literature search & corpus

MCP，converts research papers into artificial intelligence (AI) agents

ERA(Google, 2026.5.19)->scientific simulation code

Automatically create expert-level scientific software for simulation

AI X-ray Scientist(SLAC & Stanford, 2026.6.1) -> experiment execution

task: 搜索 Bragg Peak，优化电机角度与衍射峰强度

technique: LLM + MCP + Virtual(Sim)-to-Real Transfer

结果分析 Multimodal Reasoning + 在线调整 In-context Adaptation

Reference

Part II: ml in nuclear astrophysics